

CANDIDATE

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TEST

DAT232 HT25 LP1 Ordinarie tentamen - J

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CHALMERS | GÖTEBORGS UNIVERSITET

DAT230/DAT232/DIT285 Advanced Requirements Engineering

Exam

Thursday, October 30th 2025

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Allowed tools / material
- English dictionary

General information
The exam consists of three parts:
- Part 1: 40 multiple choice questions, each with one correct answer and worth 1 Point.
- Part 2: Interactive questions. Numbers within parentheses show the maximal points awarded for each question.
- Part 3: Essay questions. Maximal points can be given if:

  - The answer is correct.
  - The presentation of the answer is readable and clear.
  - The answer is given in English.

Grading
The grades on this exam are based on your total score on the questions:
0 – 39 points: Fail
40 – 55 points: 3
56 – 67 points: 4
68 – 80 points: 5

Review
Will be announced on Canvas.

Part 1: Multiple Choice Questions

1.1

Proposition	Reason
According to IEEE Definition of *Requirement* from 1990, a requirement must be explicitly written down, otherwise it is not a requirement.	According to the IEEE Definition from 1990, a Requirement is: (1) A condition or capability needed by a user to solve a problem or achieve an objective. (2) A condition or capability that must be met or possessed by a system or system component to satisfy a contract, standard, specification, or other formally imposed document. (3) A documented representation of a condition or capability as in (1) or (2).

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement. ✓
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.2

Proposition	Reason
Tacit knowledge is difficult to transfer to another person by writing it down or verbalizing it.	Developers and users may not share a common language, thus making it hard to communicate about requirements.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition. ✓
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.3

Proposition	Reason
Only real persons can be stakeholders.	A stakeholder must have a valid id with their contact information.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false. ✓

Correct. 1 of 1 marks.

1.4

Proposition	Reason
According to Lauesen, interviews are not suitable for conflict resolution.	Interviews must always be conducted with a single individuum as interviewee.

Select one alternative:

A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.

B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.

C) The proposition is a true statement, but the reason is false. ✓

D) The proposition is false, but the reason is a true statement.

E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.5

Proposition	Reason
According to Lauesen, each group that participates in a focus group must "get something".	If a stakeholder group does not get something in return, they are rarely willing to contribute to the system.

Select one alternative:

A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct. ✓

B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.

C) The proposition is a true statement, but the reason is false.

D) The proposition is false, but the reason is a true statement.

E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.6

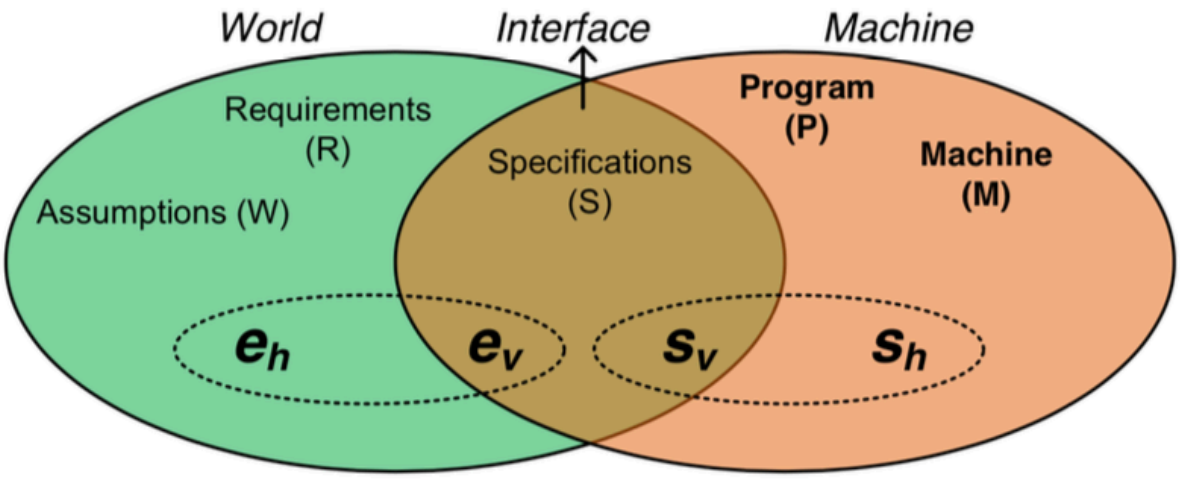
Proposition	Reason
Goal domain tracing is both useful for elicitation and validation of requirements.	Goal domain tracing allows to validate whether each task or quality factor has a purpose with respect to business goals as well as to validate that each business goal is sufficiently supported by tasks and quality factors.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Wrong. 0 of 1 marks.

1.7



Proposition	Reason
The WRSP model does support architectural decomposition.	The WRSP model defines requirements and specification based on events in the "world" and states in the "machine". These states can be used to describe the logical architecture, thus allowing to map requirements to logical components.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement. ✔
- E) Both the proposition and the reason are false. ✖

Wrong. 0 of 1 marks.

1.8

Proposition	Reason
According to Glinz, an attribute is a performance requirement, a specific quality requirement, or a constraint.	It is equally difficult to make performance requirements, specific quality requirements, and constraints measurable.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false. ✓

Correct. 1 of 1 marks.

1.9

Proposition	Reason
Lauesen discourages using requirements as part of a contract between customer and supplier.	If requirements are part of a contract between customer and supplier, changing a requirement entails changing the contract, which may involve legal teams on both sides.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement. ✓
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.10

Proposition	Reason
It is hard to ensure that a requirements specification is complete.	Completeness of a requirements specification refers to all necessary requirements being included.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Wrong. 0 of 1 marks.

1.11

Proposition	Reason
It is insufficient to define completeness of a specification in relation to the template for the specification.	Lauesen warns about template blindness, which can narrow down ones view to what the template deals with.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.12

Proposition	Reason
A feature based approach to documenting requirements facilitates an unambiguous requirements specification.	In a feature based approach, each requirements is one statement about a feature of a system.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement. ✔
- E) Both the proposition and the reason are false.

Wrong. 0 of 1 marks.

1.13

Proposition	Reason
In order to achieve a good enough requirements specification, it is important to focus on the requirements that everybody knows about.	Requirements that are well known by many stakeholders are easily misinterpreted.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false. ✔

Correct. 1 of 1 marks.

1.14

Proposition	Reason
Data expressions are suitable for specifying message protocols.	Data expressions are simplified screens with realistic data.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false. ✓
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.15

Proposition	Reason
The event list & function list approach allows to trace between product-level requirements and design-level requirements.	The event list & function list approach establishes many-to-many relationships between domain events and product events.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement. ✓
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.16

Proposition	Reason
To a degree, scenario-based approaches such as task descriptions can support completeness of functional requirements.	It is relatively easy to spot missing sub-tasks in task descriptions

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct. ✓
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.17

Proposition	Reason
Creativity techniques in requirements engineering should replace elicitation.	Stakeholders cannot provide novel requirements.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement. ✗
- E) Both the proposition and the reason are false. ✓

Wrong. 0 of 1 marks.

1.18

Proposition	Reason
Convergent thinking (to evaluate and choose the best ideas) should be avoided during Creative Problem Solving (CPS).	A core principle of CPS is to defer judgment.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement. ✔
- E) Both the proposition and the reason are false. ✖

Wrong. 0 of 1 marks.

1.19

Proposition	Reason
A taxonomy of system and software quality characteristics helps to identify relevant quality attributes.	Many established quality taxonomies such as ISO25010 and ISO25019 are incomplete.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason do[illegible]not explain the proposition. ✔
- C) The proposition is a true statement, but the reason is false. ✖
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Wrong. 0 of 1 marks.

1.20

Proposition	Reason
Functionality is not a quality characteristic.	Quality characteristics include specific quality attributes and performance, but no functional requirements.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct. ✖
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement. ✔
- E) Both the proposition and the reason are false.

Wrong. 0 of 1 marks.

1.21

Proposition	Reason
Open metric allows to rely on the suppliers expertise with respect to knowing how to measure the quality of the system.	Open metric delegates responsibility for reaching the project goals to the supplier, who is now responsible for the acceptance test of quality requirements.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false. ✔
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.22

Proposition	Reason
Planguage is a template that facilitates testable spefication of quality requirements.	Specifically, the Meter field in the Planguage template ensures testibility of requirements.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false. ✓
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.23

Proposition	Reason
Priority grouping is a simple, but often flawed approach to prioritization.	Priority grouping does not require challenging requirements against each other and may lead to too many high priority requirements.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct. ✓
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.24

Proposition	Reason
In prioritization, the ordinal scale should be favored, since it has the highest expressive power of all available scales.	The ordinal scale allows to express that requirement A is 2x as important than requirement B.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false. ✓

Correct. 1 of 1 marks.

1.25

Proposition	Reason
Horizontal traceability refers to tracing across different phases of the waterfall model.	Ramesh and Edwards define horizontal traceability as tracing between artifacts of different types (e.g. tests and requirements).

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false. ✓

Correct. 1 of 1 marks.

1.26

Proposition	Reason
Pre-requirements traceability supports validation.	Pre-requirements traceability connects requirements to their source or origin, thus establishing whether a requirement is needed.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct. ✓
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.27

Proposition	Reason
In change impact analysis, traceability can be used to generate the Candidate Impact Set.	If important tracelinks are missing, the candidate impact set my not include all artifacts of the actual impact set.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct. ✗
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition. ✓
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Wrong. 0 of 1 marks.

1.28

Proposition	Reason
GDT is a specific traceability matrix.	Traceability matrices are often shown with the same artifacts in both dimensions and an indication in a "cell" if the artifacts of this cell's column and row are connected.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason do✔not explain the proposition.
- C) The proposition is a true statement, but the reason is false. ✖
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Wrong. 0 of 1 marks.

1.29

Proposition	Reason
A graphical model that connects business goal 1 with task description 7 is not a TIM.	A TIM is the metamodel that defines which tracelinks between artifacts are allowed and which tracelinks between artifacts are required.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the propos✔n is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false. ✖

Wrong. 0 of 1 marks.

1.30

Proposition	Reason
Even outdated tracelinks are better than no tracelinks.	Outdated tracelinks create additional work, as irrelevant artifacts must be considered when relying on traceability. At the same time, relevant artifacts may be overlooked, since a tracelink has not yet been set. In particular the latter can be critical in many use cases.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement. ✓
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.31

Proposition	Reason
A CRUD matrix supports verification of requirements.	Verification can be expressed as the question "Do the requirements fulfil the stakeholder needs/goals" and a CRUD matrix allows to trace from data requirements to goals.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false. ✓
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.32

Proposition	Reason
Inspections can provide both validation and verification of requirements.	Inspections are a systematic method for manually reading through specifications.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Wrong. 0 of 1 marks.

1.33

Proposition	Reason
Usability tests is a good approach for requirements verification.	During a usability check, the facilitator will guide the user through checking the consistency of requirements.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.34

Proposition	Reason
In agile software development, development teams will have to do "Just-in-time" RE.	The only other option would be to fully analyse the requirements before the team starts to develop, effectively establishing a waterfall process.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the propos ✔ ı is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false. ✖

Wrong. 0 of 1 marks.

1.35

Proposition	Reason
A user story is more than just the text, it should also include confirmation, i.e. describing its acceptance test.	The most common template for a user story (As a <role> I want <feature> so that <value>) ensures that a user story is testable.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false. ✔
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.36

Proposition	Reason
In large-scale agile approaches, non-functional requirements should not be managed on team level.	Non-functional requirements are usually cross-cutting, thus, they affect multiple components, functional requirements, and teams.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct. ✓
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.37

Proposition	Reason
Many modern vehicles (not only those with self-driving capabilities) take pictures and videos of their surroundings, especially of humans that approach them while they are parked. This can be considered as an ethical problem.	Imagine a school child approaching a parked vehicle. The vehicle taking pictures and sharing them with an unknown group of recipients can be considered a violation of the EU ethical guideline of Nonmaleficience.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false. ✓
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.38

Proposition	Reason
During requirements analysis, anonymisation is no longer a concern.	Anonymisation (defined as a component of confidentiality involving not revealing any data that can be used to identify subjects) should happen during elictation.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement. ✓
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.39

Proposition	Reason
Ambiguous requirements can cause ethical problems.	Ambiguity in requirements often leads to ambiguous or biased interpretation, however, biases and fairness are important ethical consideration.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct. ✓
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

1.40

Proposition	Reason
Optimizing for overall safety may still disadvantage certain groups (e.g., cyclists, children, or pedestrians in low-visibility conditions).	Without additional effort for these groups, the overall safety strategy may not sufficiently represent these groups because a) they may be underrepresented in the data (particular in low-visibility conditions) and b) they are more vulnerable than other traffic participants.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

Part 2: Interactive questions

2.1

Please match the requirements in the table below with the most suitable alternative level.

	Goal-level	Domain-level	Product-level	Design-level
The menu points and buttons shall work as described in appendix yy.				✓
The product shall support the user registration task.		✓		
The product shall have a function for retrieving items based on specified keywords.			✓	
The product shall support estimations that match actual scores within a standard deviation of 1%.	✓	✗		
The product screens shall look as shown in appendix xx.				✓
For all fields, the user shall be able to select the value from a drop-down list.				✓
The system shall have a mini keyboard with start/stop button.			✗	✓
The product shall have a function for storing new items based on their associated keywords.			✓	
The product shall handle the booking task.		✓		
The system shall allow efficient usage with one hand only.		✗	✓	

Partially Correct. 7 of 10 marks.

2.2 Consider the following requirements in a bespoke project that aims to build a tailor-made solution. Judge whether they mainly pose a risk for the customer or the supplier and select the most valid reason.

R1 At most 1 of 5 novices shall encounter critical problems during tasks Q and R. At most 5 medium problems shall be encountered in total.

This requirement mainly poses a risk for the Supplier ✓ (Customer, Supplier),

since It cannot be known whether this requirement is feasible. ✓ (It cannot be known whether this requirement is feasible., The requirements style does not allow for usability tests, The customer might not accept the solution even if the test is successful, The solution might not work for the customer even if the requirement is fulfilled)

R2 Novice users shall perform tasks Q and R in 15 minutes. Experienced users shall perform tasks Q, R, and S in 2 minutes.

This requirement mainly poses a risk for the Supplier ✓ (Customer, Supplier),

since The requirement cannot be assessed early in the project ✓ (The product may not be usable even if the test is successful, The proper time limit is hard to define, The requirements style does not allow for usability tests, The requirement cannot be assessed early in the project)

R3 Recording breakfast shall be possible with 5 keystrokes per guest. No mouse.

This requirement mainly poses a risk for the Customer ✓ (Supplier, Customer),

since The solution might not work for the customer even if the requirement is fulfilled ✓ (The requirement cannot be assessed early in the project, The requirements style does not allow for usability tests, It cannot be known whether this requirement is feasible, The solution might not work for the customer even if the requirement is fulfilled)

R4 System shall use screen pictures in app. xx, buttons work as app. yy

This requirement mainly poses a risk for the Supplier ✗ (Supplier, Customer),

since It cannot be known whether this requirement is feasible ✗ (The requirements style

does not allow for usability tests, It cannot be known whether this requirement is feasible, The requirement cannot be assessed early in the project, The solution might not work for the customer even if the requirement is fulfilled)

R5 The system shall follow style guide zz. Menus shall have at most three levels.

This requirement mainly poses a risk for the Customer ✓ (Supplier, Customer),

since The solution might not work for the customer even if the requirement is fulfilled ✓ (The requirement cannot be assessed early in the project, The solution might not work for the customer even if the requirement is fulfilled, The requirements style does not allow for usability tests, It cannot be known whether this requirement is feasible)

Partially Correct. 8 of 10 marks.

Part 3: Essay questions

For each topic below, write an essay that includes all of the listed concepts within the given maximum number of pages. The essays are graded based on

(a) how well the topic is described using the concepts in the list under each topic (thus judging the general flow and argumentation), and

(b) how well the concepts in the list are described and/or exemplified in the essay (thus, points are given for each concept that is mentioned and explained in sufficient depth).

3.1 Topic: Elicitation **(10p)**

List of concepts: 3 examples of elicitation techniques (including how they are used, when to use them, when not to use them, if and how they should be combined), stakeholder, negotiation of conflicting requirements

Write your answer in the box below. Changes are saved automatically.

Elicitation is the process of searching, discovering and elaborating requirements by applying specific techniques that each have their own strengths and weaknesses. Choosing a suitable technique is done based on the context. The first thing to do is exploring the domain and identifying the stakeholders: (groups of) people that will be affected by the product. They are necessary to find requirements. By constructing a stakeholder map we can get an overview of them. For each stakeholder it should include: name or role if it is a group, (representative) contact information, sentiment and impact on the product. Additional information that is useful: what can the stakeholder contribute to the system, what risks do they see and how would they solve those?

Depending on the phase of the project different techniques are better suited. During the early phase, observation is good to get an overview of present processes and problems because you observe how users perform their tasks. However, it is time consuming and you might miss rare but critical problems.

During the mid phase of the project you already have a basic understanding of the domain and necessary requirements, making it suitable to use a creative approach to discover novel and useful ideas. Brainstorming allows you to come up with a lot of ideas. First you just write everything down and don't criticize or evaluate any of them. After you came up with enough ideas they are evaluated and the best ones can be chosen to add as new requirements.

During the last phase it is good to test the product, and even better to do so before it has been (fully) implemented. Prototyping provides a solution by creating either a formal example or a mockup of the final system. Usability testing lets real user perform real task using the prototype. Discovered issues can be resolved early on in the development process which is less costly than having to redesign an implemented system.

It is possible that during this process some requirements are conflicting with each other. This can for example happen with specific quality requirements such as usability and security. A login function is more secure but less good from the perspective of a user as it can get tedious. In this case it is important to gather more information from the impacted stakeholders. If tough decisions have to be made it is good to prioritize requirements and choose based on the results. It is important to choose a suited criteria to prioritize on (e.g. budget, time to implement, need by the user) and choose an appropriate technique (e.g. 100$ prioritization where you have a fixed budget and need to assign money to requirements based on how important they are).

Words: 450

Answered.

3.2 Checking and Validation **(10p)**

List of concepts:
- complete,
- unambiguous,
- consistent,
- modifiable,
- verifiable

Write your answer in the box below. Changes are saved automatically.

Checking and validation is used to ensure that all work performed by requirement engineering activities is useful and correct. Through validation you ensure that all requirements reflect the need of a user, and as such can be traced back to a goal or domain-level requirement. Verification can be done to check if requirements are correct with respect to what we know. Pre-determined quality criteria define what a "good" specification should meet. Some of these are:

A complete specification means that it addresses all needs by the user. There are no needs that are unfulfilled. Validation techniques are used to check this.

An unambiguous specification is clear: all different stakeholder(s) (groups) agree on its meaning.

Consistency: there are no conflicts within the specification. This can happen between different requirements or abstraction-levels. E.g. the functional and data requirements don't match, there is a retrieve function for data that does not exist.

Modifiable: it should be easy to modify the specification.

Verifiable: each requirement can be verified or tested to ensure it is met. This implies that all requirements are atomic, because we cannot easily tell whether a non-atomic requirement is met.

One way of checking that the above criteria are fulfilled is through inspection. The document is manually read, with or without a predetermined checklist. The latter is useful when specific areas need to be inspected. Possible things to check for in a checklist:

1. Content checks ensure that all necessary topics are present in a specification. A few examples of these are business goals, functional requirements or data requirements. These checks address the "complete" criteria.

2. Structure checks do not evaluate the content itself, but rather how the specification is constructed. Examples: Does each requirement have a unique id? Is each requirement verifiable? Does each requirement have a purpose?

3. Consistency checks control that all parts of the specification agree with each other.

Reviews are done with multiple stakeholders present. First everyone reads the specification in isolation, and afterwards you meet to discuss noted issues. You can do a general review or focus on more specific parts such as traceability to ensure all user needs are met or a risk analysis where every party mentions risks they found from their perspective.

A third approach to evaluate is through tests. These focus on the product itself rather than use the specification. Usability tests involve a user performing realistic tasks on the product (can also be mockup or prototype). A logger and facilitator are also present. Discovering issues early-on in the development cycle saves time and money because modifications are easier to do. Another test approach is heuristic evaluation where an expert assesses the product based on heuristics. An advantage is that no real user is required, but a lot of issues are often not found, and other raised issues might not even be a problem in general.

Words: 475

Answered.